



Lida Kanari

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- From Athens, Greece
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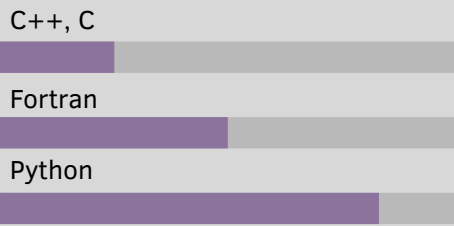
About me

I am a trained mathematician and physicist, who works in applications of mathematics to neuroscience and biology. My current research bridges the worlds of algebraic topology and computational modeling to understand how neurons give rise to complex brain systems.

Interests

- > Complex Systems
- > Topological Data Analysis
- > Computational Topology
- > Stochastic Processes
- > Machine Learning
- > Neuroscience

Programming



Languages

Greek, English, French

Research experience

2025–now	Research Fellow & Lecturer <i>NeuroTopology group</i>	University of Oxford, UK
2022–2025	Group Leader & Lecturer <i>Neuromathematics group</i>	EPFL, Lausanne
2019–2022	Group Leader <i>Neuromathematics group</i>	EPFL, Lausanne
2018–2019	Post-Doc Computational Neuroscience <i>Neuronal morphologies</i>	EPFL, Lausanne
2011–2012	Research assistant Biomechanics lab	NTUA, Greece
2009–2010	Summer School Internship Analysis of particle detection	CERN, Geneva

Education

2012–2018	Ph.D. Computational Neuroscience <i>Neuronal morphologies: the shapes of thoughts.</i>	EPFL, Lausanne
2011–2012	M.Sc. Computational Mechanics Majoring in Fluid mechanics and Biomechanics	NTUA, Greece
2006–2011	M.Sc. Applied Mathematics and Physics Majoring in Theoretical and Particle Physics	NTUA, Greece

Awards

2018, 2017	OCNS	Travel and Poster awards
2011-2012	Limmat Stiftung Scholarship MSc, Computational Mechanics State Foundation of Fellowships, Greece MSc, Computational Mechanics	Highest GPA State Fellowship
2006-2011	State Foundation of Fellowships, Greece MSc, Applied Mathematics and Physics	Highest GPA
2009-2010	International Mathematical Competitions	Competition Award
2003-2005	Hellenic Mathematical Society	Competition Awards

Funding

2024	Algebraic topology bridging the gap between single neurons and networks Career Development Grant	UKRI - MRC 1.4M £
2024	Role of top-down contextual representations and inhibitory sub-populations in neocortical mismatch reponses coCP with Eilif Muller, Catalyseur-Fondation Courtois	100K
2023	S^1 -persistence modules and Persistent Homology Transform coPI with Kate Turner and Barbara Giunti SAGES	14K
2022	Mouse Neuronal Morphologies in Health and Disease: Reconstruction, Analysis and Modelling Workshop organization grant	30K

Patents

Title : Synthesis of branching morphologies
Application No : 20190100470
Inventors : Lida Kanari, Kathryn Hess Bellwald, Henry Markram

Teaching experience

2025–now	Machine Learning by Dr. L. Kanari	University of Oxford, Oxford
2022–2023	Mathematical Methods for Neuroscience by Dr. L. Kanari	EPFL, Lausanne
2015–2022	Simulation Neuroscience by Profs. H. Markram & I. Segev	MOOC in EdX, online
2016–2017	Symmetry and Conservation in the Cell by Dr. J. Shillcock	EPFL, Lausanne
2012–2014	Analysis II, Advanced Analysis II & Numerical Analysis by Profs R. Hans-jörg & F. Margot & S. Deparis	EPFL, Lausanne

Project supervision and mentoring

2023-2025	Darshan Mandge, Anil Tuncel, Ilkan Kilic, Aurelien Jaquier Project: <i>Modeling of dendritic electrophysiology</i>	Software engineers, EPFL
2020-2025	Adrien Berchet, Remy Petkantchin, Valerii Souchoroukov Project: <i>Axon synthesis</i>	Postdoctoral researchers, EPFL
2019-2025	Ying Shi Project: <i>Neuronal reconstructions</i>	Reconstruction specialist, EPFL
2019-2025	Alexis Arnaudon Project: <i>Morpho-electrical properties of neurons</i>	Postdoctoral researcher, EPFL
2019-2022	Stanislav Schmidt Project: <i>Machine learning of neuronal morphologies</i>	Software engineer, EPFL

Student supervision

2018-2025	Alexia Stéphanie Liviana Paratte, Romane Cornali, Thomas Negrello Alain Walter, Jelena Banjac, Marc Fersztand, Abel Adary, Sébastien Morand	Master thesis, EPFL
2020-2025	Jan Zraggen, Mariia Vidmuk, Ananda Sophie Wolz, Christina Kapatsori, Killian Rigaux, Camille Frayssinhes, Dmitriev Daniil	Master project, EPFL
2014-2024	Zhixin Wang, Jeanne Sentenac, Rebecca Elyanow	Internship, EPFL

Selected Outreach

2025	From neurons to Networks	GEOTOP seminar
2023 - 2022	Decoding neuronal variability, Building neurons with mathematics, Pioneering Algebraic Topology, The real neuron challenge	EPFL news
2019	A solution to a century old neuroscience problem	EPFL news

Boards

2026-now	Associate Editor in <i>Brain Informatics</i>	Springer Nature
2026-now	Fellow of the <i>Young Academy of Europe</i>	YAE
2026-now	Member of <i>ENCODE</i>	AI for Science

Reviewer

Nature, Nature Neurosc, Nature Comms, Nature Biotech, Plos Comp Bio, Cosyne
Journal of mathematical biology, IEEE transactions, Research in Mathematics

Publications

- *2025* Of mice and men: Dendritic architecture differentiates human from mouse neuronal networks. Kanari L. et al. *iScience*
- 2025 Breakdown and repair of metabolism in the aging brain. Shichkova, P., Coggan, J.S., Kanari, L. et al. *Front Sci*
- *2025* Generating brain-wide connectome using synthetic axonal morphologies. Petkantchin R., Berchet A., Peng H., Markram H., and Kanari L. *Nat Commun*
- 2025 Computational Generation of Long-range Axonal Morphologies. Berchet, A., Petkantchin, R., Markram, H. and Kanari L. *Neuroinformatics*
- 2024 Community-based Reconstruction and Simulation of a Full-scale Model of Region CA1 of Rat Hippocampus. Romani, A. et al. *Plos Biology*
- 2024 BlueCelluLab: Biologically Detailed Neural Network Experimentation. Tuncel A. et al. *JOSS*
- 2024 Modeling and Simulation of Neocortical Micro- and Mesocircuitry. Part I: Anatomy. Reimann M. et al. *elife*
- 2023 Controlling morpho-electrophysiological variability of neurons with detailed biophysical models. Arnaudon A. et al. *iScience*
- 2023 From Trees to Barcodes and Back Again II: Combinatorial and Probabilistic Aspects of a Topological Inverse Problem. Curry, J. at el. *Computational Geometry.*
- 2023 Strong and reliable synaptic communication between pyramidal neurons in adult human cerebral cortex. Hunt, S. et al. *Cerebral cortex*
- *2022* Computational synthesis of cortical dendritic morphologies. Kanari L. et al. *Cell Reports*
- *2022* A tool for mapping microglial morphology, morphOMICS, reveals brain-region and sex-dependent phenotypes. Colombo G. et al. *Nat Neurosc*
- 2021 Digital Reconstruction of the Neuro-Glia-Vascular Architecture. Zisis E. et al. *Cerebral Cortex*
- 2020* From trees to barcodes and back again: theoretical and statistical perspectives. Kanari L., Garin A., and Hess K. *Algorithms*
- 2019 Objective Morphological Classification of Neocortical Pyramidal Cells. Kanari L. et al. *Cerebral Cortex*
- *2018* A Topological Representation of Branching Neuronal Morphologies. Kanari L. et al. *Neuroinformatics*
- 2017 Comprehensive Morpho-Electrotonic Analysis Shows 2 Distinct Classes of L2 and L3 Pyramidal Neurons in Human Temporal Cortex. Deitcher Y. et al. *Cerebral Cortex*
- 2016 Framework for efficient synthesis of spatially embedded morphologies. Vanherpe L., Kanari L. et al. *Phys. Rev. E*
- 2016 In situ synthesis and simulation of polydisperse amphiphilic membranes. Vanherpe L., Kanari L. et al. *IJAESAM*
- 2015 Reconstruction and Simulation of Neocortical Microcircuitry. Markram H. et al. *Cell*
- 2015 The neocortical microcircuit collaboration portal: a resource for rat somatosensory cortex. Ramaswamy S. et al. *Front. Neural Circuits*

Preprints

- 2025 Quantifying neuronal differentiation using temporal topological persistence. Rigaux K., Castro A.F. and Kanari L. *bioarxiv, under review*
- 2025 Neuron morphological physicality and variability define the non-random structure of connectivity. Reimann M.W., Egas-Santander D., Lida Kanari L. and Barros-Zulaica N. *bioarxiv, under review*
- 2024 A quantitative spatial atlas of transcriptomic, morphological, and electrophysiological cell type densities in the mouse brain. Verasztó C., Roussel Y., Kanari L., Markram H. and Keller D. *bioarxiv, accepted*
- 2024 Deep learning for classifying neuronal morphologies: combining topological data analysis and graph neural networks. Kanari L. et al. *bioarxiv, under review*
- 2024 Decomposing the Persistent Homology Transform of Star-Shaped Objects. Arya S., Giunti B., Hickok A., Kanari L., McGuire S. and Turner K. *arxiv, accepted*

Invited speaker - Selected Conferences

2025 AI	Erlangen - TDA Seminar October <i>From Neurons to Networks: Understanding the Brain through Algebraic Topology</i>	London, UK
2025	Queen Mary, University of London <i>From neurons to complex human networks using algebraic topology</i>	London, UK
2025 CNS	Tutorial <i>Open Brain Institute: From single neurons to complex networks</i>	Florence, Italy
2025 BrainNet	<i>From single neurons to complex networks using algebraic topology</i>	Stockholm, Sweden
2024 Oxford	<i>From single neurons to complex human networks using algebraic topology</i>	Oxford, UK
2024 EPFL	Applied Machine Learning Days <i>From single neurons to complex human networks</i>	Lausanne, Switzerland
2023 ICERM	Topology and Geometry in Neuroscience <i>From single neurons to complex networks using algebraic topology</i> invited by Carina Curto	Providence, US
2023	Microstructure by the Lake <i>From single neurons to complex networks using algebraic topology</i> invited by Erick J Canales-Rodriguez	Lausanne, Switzerland
2023	Form and Function of Complex Systems <i>From single neurons to complex networks using algebraic topology</i> invited by Gabriele Manoli	Lausanne, Switzerland
2023 WCT	Women in Computational Topology <i>Persistence Homology Transform</i>	Lausanne, Switzerland
2022	VU Amsterdam Math Neuro workshop <i>Algebraic topology insights on mouse and human neuronal morphologies</i> invited by Daniele Avitabile	VU, Amsterdam
2022 BICCN	Morphology Working Group <i>Computational synthesis of cortical dendritic morphologies</i> invited by Giorgio Ascoli	Online
2021 MSRI	Hot Topics: Topological Insights in Neuroscience <i>Topological insights on neuronal morphologies</i>	Online
2020 NeurIPS	Topological Data Analysis and Beyond <i>Topological insights on neuronal morphologies</i>	Online
2020 Cosyne	Neural networks that are neuronal networks: Considerations of neuron morphology in circuit computations <i>Topological synthesis</i>	Denver, US
2019 CNS	CNS <i>Topological synthesis</i> invited by Hermann Cuntz	Seattle, US
2017 ELSC	Edmond and Lily Safra Center for Brain Sciences retreat <i>Topological representation of neuronal morphologies.</i> invited by Idan Segev	Ein Gedi, Israel
2017 AAT	Applied Algebraic Topology <i>Topological representation of neuronal morphologies.</i> invited by Kathryn Hess	Sapporo, Japan
2016 BNH	BigNeuron Hackathon <i>Analysis, validation and synthesis of neuronal morphologies.</i> invited by Hanchuan Peng	London, UK
2016 ASMP	Applications and Statistics of Multidimensional Persistence <i>Topological representation of branching neuronal morphologies.</i> invited by Scolamiero Martina	Lausanne, Switzerland
2015 BIH	International Conference on Brain Informatics and Health <i>Automated reconstruction, analysis, classification and synthesis of neuronal morphologies.</i> invited by Hanchuan Peng	London, UK